Distilling and Reinforcing Macedonian Poetic Style Transfer: SFT and GRPO on a 4B Student

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Abstract

Gemma 4 31B, given a heavily scaffolded prompt (author style profile, vocabulary palette, exemplar lines, structural targets), rewrites a Macedonian poem in a named poet's style. We distil this into Qwen3-4B given only the source poem and the target author's name: LoRA supervised fine-tuning (SFT) on 1,146 teacher outputs, then Group Relative Policy Optimization (GRPO) against a reward built from the project's stylometric profiles, line-level content similarity and gates against copying, repetition and malformed output. SFT teaches the task format completely but collapses ten voices into one dominated by the corpus's two largest authors. GRPO-v1 raises the reward from 0.254 to 0.406, removes SFT's repetition and doubles content preservation, but lowers author identification and learns to copy single source lines. GRPO-v2, which reweights the classifier term and steepens the copy gate, keeps identification near the SFT level only by holding the policy on the two largest authors. For eight of the ten targets neither student is ever identified as the requested author, which leaves GRPO's group-relative update nothing to amplify. All results are on the validation split; the test-split run is in progress.

1 Setup

Task. *source poem* + *target author's name* $\rightarrow$ *new poem* in the target's voice, in Macedonian Cyrillic. The teacher (Gemma 4 31B behind a rate-limited API, about three minutes per poem including obligatory reasoning) needs the whole stylometric pipeline at inference time; the student needs none of it.

Data. `make_splits.py` splits the 1,239 poems of the 38 profiled authors 80/10/10 per author (seed 42) into 987 train, 126 val and 126 test poems; the unit is the poem, since (author, title) is not unique in the corpus. The ten target authors are the ten with the most poems: Блаже Конески, Пијан Славеј, Ефтим Клетников, Ристо Лазаров, Веле Смилевски, Братислав Ташковски, Влада Урошевиќ, Адем Гајтани, Славко Јаневски and Катица Ќулавкова. The teacher wrote 1,146 transfers for (train poem, target) pairs, 108–120 per target, reusing the same 130 source poems across targets so that the data contrasts authors on identical content.

Evaluation. Every system is generated for the same 468 val pairs (50 val poems $\times$ 10 targets, self pairs skipped), with the same prompt, per-item seeds and plain sampling ($T = 0.7$, top-$p = 0.9$, top-$k = 20$, 512 new tokens), on one A100. The teacher was run on the same 468 pairs as a reference. Metrics (`style_metrics.py`) are all higher-is-better and shared with the GRPO reward:

- *degeneracy*: distinct-n, line uniqueness; an output is *degenerate* if distinct-1 < 0.40 or line uniqueness < 0.80;
- *copying*: *line novelty* (share of lines not verbatim from the source) and *copy novelty* (share not near-copies; §**??**);
- *content*: content-word recall from the source;

- *target style*: structural fit, TF-IDF palette hit rate, and two author classifiers. The reward classifier (`classifier.pkl`, character 2–4-grams, linear SVM) was fitted on the whole corpus, so it memorises real poems (top-1 1.000 against 0.883 under 5-fold CV), and it is a GRPO reward term. The *judge* is evaluation-only and differs on every axis: train poems only, word uni+bigrams, logistic regression. Its top-1 on unseen real poems of the ten targets is 0.663, the ceiling for a perfect transfer.

2 Supervised fine-tuning

The student sees a minimal chat: a one-sentence system instruction, the source poem and the target's name, and the teacher's poem as the answer. None of the teacher's scaffolding is in the prompt; whatever it knows about an author it learns from about 115 teacher poems. One module (`sft_data.py`) renders the prompt for training, evaluation and GRPO, so they cannot drift. LoRA (rank 16, α 32, dropout 0.05) on all attention and MLP projections of `Qwen/Qwen3-4B`; 3 epochs, effective batch 16, learning rate 2×10^{-4} (cosine), max length 1024, bf16, loss on the assistant turn only. The run took 210 steps and 15 minutes on one A100. Loss fell from 2.87 to 1.72 in the first epoch and to 1.34 by the third, with steps at the epoch boundaries that suggest partial memorisation of the training poems. 71 examples (6.2 %) are longer than 1024 tokens and lost their end-of-turn token; SFT was not retrained, to keep both GRPO runs' start policy fixed.

Result. Against the untuned base (Table **??**), SFT does the task: the base model mostly reproduces the source under a markdown heading (line novelty 0.224, markup-free in 59 % of outputs), while SFT writes original lines (1.000), never adds markup or commentary, fits the target's structure (0.846 vs. 0.680), and is identified as the target 17 times more often (classifier top-1 0.158 vs. 0.009; judge 0.158 vs. 0.064). The base model's higher content recall (0.901) and diversity come from the copied human text.

Two deficits. *Style collapse*: the aggregate is carried by Конески and Пијан Славеј, the corpus's two largest authors; the other eight targets are identified 0–15 % of the time (Table **??**), and 36 % of all outputs are read as Конески whatever the target. The training data is balanced across targets, so the imbalance is not one of example counts. Pooled vocabulary at a matched 30,000-token budget is 3,087 word types, against 5,376 for the teacher and 9,060 for real poems. *Repetition*: 30.1 % of outputs are degenerate. A decoding ablation separates the two: a repetition penalty of 1.1 removes degenerate output entirely (line uniqueness 0.999) but costs a third of content recall, and on 56 pairs no decoding setting moves author identification by more than noise. Repetition is partly decoding; the collapse is in the policy, and is what GRPO is for.

3 GRPO

GRPO samples $G = 4$ completions per prompt, scores each, and normalises the rewards within the group, so the update moves probability towards the group's above-average samples. The reward sees only the generated poem, the source poem and the target author — never the teacher's output, and no val or test poem. With a LoRA policy, TRL's KL reference is the policy with its adapter switched off; the SFT adapter is therefore merged into the base weights and a fresh LoRA trained on top, so that the start policy and the KL reference are both exactly the SFT model.

3.1 Reward

Style. Every feature is scored against the target author's *per-poem* distribution over their train poems. A scalar feature scores $s_f = \exp(-\frac{1}{2} z_f^2)$ with $z_f = (x_f - \mu_{a,f}) / \max(\sigma_{a,f}, 0.5\,\sigma_{\text{corpus},f})$; POS-bigram and rhyme distributions are scored by cosine to the author's distribution, one-sided against the author's own leave-one-out

self-similarity; the reward classifier's target probability is used as is; palette hits are divided by the author's 90th-percentile rate on real poems. R_{style} is the weighted mean of the groups:

Group	Features	v1	v2
POS profile	share of NOUN, VERB, ADJ, ADV, PROPN	1.0	1.0
lexical diversity	MATTR over lemmas	1.0	1.0
structure	tokens/line, lines, stanzas, lines/stanza	1.0	1.0
POS bigrams	bigram distribution	1.0	1.0
classifier	reward classifier's target probability	1.0	**5.6**
rhyme, TF-IDF	rhyme endings; palette hits	0.5	0.5

Content. A line-level, BERTScore-style F1 against the source over multilingual sentence embeddings: each source line's best match among the generated lines (recall) and the reverse (precision). Because every line takes its best match, unrelated poems already score $b = 0.553$, so $R_{\text{content}} = \text{clip}\big((F_1 - b)/(1 - b), 0, 1\big)$.

Gates. Behaviours cheaper than doing the task are multiplied out rather than traded off. *Copying*: R_{content} is multiplied by copy novelty, where a line of $\geq$ 3 tokens is a copy if $\geq$ 75 % of its word types occur in one source line (containment, because Jaccard lets a one-word edit of a 5-token line through); v2 uses copy novelty^3. *Repetition*: R_{style} is multiplied by line uniqueness. *Format*: both are multiplied by `is_cyrillic` · `no_chat_markup` · `is_nonempty`. The total is $R = \alpha R^g_{\text{style}} + \beta R^g_{\text{content}}$.

Weights from within-group spread. Under group-normalised advantages a term steers the update in proportion to its spread *within* a group of samples, not its mean. Sampling four SFT completions per train prompt, the gated style term's within-group standard deviation is 0.113 and the content term's 0.042, so training uses $\alpha = 0.3, \beta = 0.7$. Within style, the classifier's is 0.051 against 0.26–0.31 for lexical diversity and POS bigrams. Before training, the reward was checked on hand-built candidates: a real poem by the target scores highest on style, and verbatim and near-copies of the source lose their whole content reward.

Training. 500 steps, 8 prompts × 4 samples per step (16,000 scored completions), LoRA rank 16, learning rate 10^{-5}, temperature 0.9, KL coefficient 0.04, 512-token completions with truncated ones masked, bf16. Each run took about 8 hours on one A100 (v1 7.9 h, v2 8.3 h), dominated by generation and by POS-tagging every sample for the reward. v2 differs from v1 only in the two reward changes.

4 Results

4.1 GRPO v1

The reward rose and was not gamed on the gated axes. v1 beats its start policy on 94 % of the paired items. The content term more than doubles (0.123 → 0.302), and degenerate outputs fall from 30.1 % to 5.3 % under plain sampling — matching real poems, from the policy rather than a decoding control, and with higher rather than lower content recall. The style gain came from the groups with the largest within-group spread: over training, lexical diversity rose 0.49 → 0.71 and POS bigrams 0.64 → 0.84, while the classifier term *fell*, 0.102 → 0.078 (Figure **??**).

Author identification went down. Classifier top-1 fell from 0.158 to 0.079, mostly on Конески and Славеј; the share of outputs read as Конески fell from 36 % to 22 %, so v1 partly undid the collapse onto the majority voice but did not replace it with the requested ones. The judge falls less (0.158 → 0.130): a larger loss on the rewarded classifier than on an independent one is the opposite of gaming, but neither shows a gain.

Metric	Base Qwen3-4B	SFT (merged)	GRPO v1	GRPO v2	Gemma teacher	Real poems
Reward ($\alpha = \beta = 0.5$, neutral weights)						
R^g_{style} / R^g_{content}	.272/.074	.384/.123	.511/**.302**	.414/.285	**.686**/.105	—
R	.173	.254	**.406**	.350	.395	—
Task and copying						
line novelty	.224	1.000	.998	.999	1.000	—
outputs with a copied line (lower)	98.5%	8.1%	44.4%	33.8%	1.9%	—
content-word recall	.901[§]	.174	**.338**	.270	.159	—
Degeneracy						
line uniqueness	.895	.900	.971	.920	**.998**	.969
degenerate outputs (lower)	18.6%	30.1%	5.3%	25.4%	**0.2%**	6.6%
Target style						
classifier top-1 = target	.009	.158	.079	.135	**.641**	.883[†]
judge top-1 = target	.064	.158	.130	.150	**.506**	.663[†]
share read as Конески (classifier)	8.5%	36.3%	21.8%	35.9%	18.8%	—
structural fit	.680	.846	.831	.820	**.950**	.768

Table 1: All systems on the 468 val pairs. Bold: best model output per row. [§] inflated by verbatim copying. [†] held-out ceilings (5-fold CV; unseen val/test poems for the judge).

Partial copying. Outputs with at least one near-copied source line rose from 8 % to 44 %. The gate charges one copied line in twenty only 5 % of the content reward, which a gain in line-level F1 can outweigh; outputs with a copied line have content recall 0.382, those without 0.302. Checkpoint 300 confirms that later steps trade along this frontier: lower R (0.381) and recall (0.306), less copying (39 %), same identification.

4.2 GRPO v2

v2 targets both failures: the classifier group is weighted by $0.288/0.051 \approx 5.6$, giving it the same pull on the advantage as lexical diversity and POS bigrams, and the copy gate is cubed, so one copied line in twenty costs 14 % instead of 5 %.

The classifier was not gamed. The judge moves with the reward classifier, by less (classifier 0.079 → 0.135, judge 0.130 → 0.150), and agrees on 57 % of the outputs the reward classifier attributes to the target, against 62 % for SFT and 64 % for the teacher.

It held the majority voices rather than learning new ones. During training v2's classifier term stays flat (0.106 → 0.103) where v1's fell: the weight stopped the policy drifting away from the voices it had, but did not move it towards the requested ones. Almost all of v2's difference from v1 is on Конески (0.404 → 0.638) and Славеј (0.186 → 0.442), and 36 % of outputs are again read as Конески, the SFT level. The other eight targets stay at 0–9 % on both classifiers. Neither GRPO run beats SFT on author identification by either measure.

Costs. With the classifier at 5.6, the share of the style reward held by lexical diversity and POS bigrams falls from 2/6 to 2/10.6; they rise far less in training (0.48 → 0.56 and 0.63 → 0.72) and degenerate outputs return to 25 %. The cubed gate cut copying (44 % → 34 % of outputs) and with it content recall (0.338 → 0.270). By the neutral reward v2 (0.350) is below v1 (0.406), and above it on only 25 % of items.

	classifier top-1				judge top-1			
Target author	SFT	v1	v2	teacher	SFT	v1	v2	teacher
Блаже Конески	.745	.404	.638	.979	.553	.447	.660	.979
Пијан Славеј	.372	.186	.442	.651	.465	.442	.419	.698
Ефтим Клетников	.149	.021	.043	.660	.383	.298	.277	.872
Веле Смилевски	.125	.062	.083	.938	.125	.042	.042	.812
Влада Урошевиќ	.021	.021	.000	.787	.021	.064	.043	.681
Адем Гајтани	.085	.000	.021	.319	.000	.021	.000	.404
Ристо Лазаров	.064	.043	.085	.745	.021	.021	.085	.255
Славко Јаневски	.000	.020	.020	.160	.020	.000	.000	.240
Братислав Ташковски	.022	.022	.022	.609	.022	.000	.000	.109
Катица Ќулавкова	.022	.022	.022	.587	.000	.000	.000	.022

Table 2: Author identification per requested target (val pairs).

4.3 Why reweighting cannot fix the collapse

GRPO's update is driven by differences *within* a group of four samples. For eight of the ten targets the student's samples are almost never read as the target by either classifier, before or after GRPO, so all four score near zero on that term and the normalised advantage has nothing to amplify however large its weight. The only targets where the classifier term varies within a group are those the student already writes like, and that is where both runs moved. The teacher shows the voices are reachable — judge top-1 0.506, 76 % of the real-poem ceiling, with eight of the ten targets at 0.24 or above — but that capacity did not survive distillation into a 4B student from 1,146 examples. For GRPO to help, the student's SFT distribution must already put some probability on the minority voices.

The teacher also shows what the reward does not measure. Its R (0.395) is below v1's (0.406): it wins style by a wide margin (0.686 vs. 0.511) but writes a new poem on the source's theme rather than carrying its lines over, and the line-level content reward scores that barely above an unrelated poem.

5 Status and next steps

- **Test split.** All numbers above are on the val split, which was also used to make decisions (decoding, v2's reward). The final evaluation on the untouched test split — base, SFT, GRPO v1 and v2 on 466 (test poem, target) pairs with the same protocol — is running (`slurm/eval_test.slurm`) and will replace these as the reported figures.
- **Second student.** The same SFT and GRPO-v2 pipeline is being run with Gemma 4 E4B, a student of matched size from the teacher's own family, to separate properties of Qwen3-4B from properties of the method. It will be reported, and compared with these results, separately.
- **Human evaluation.** A blind rating by native speakers (language, coherence, poetic quality, faithfulness to the source, author attribution, usability) is planned, with the target's own closest real poem as a retrieval baseline.

Limitations. One seed per run, so differences of a few points (e.g. classifier top-1 0.079 vs. 0.088 between v1's checkpoints) are within noise. The reward classifier and TF-IDF palettes were fitted on the whole corpus, a leak path the judge avoids. SFT had no held-out loss, so the number of epochs was not validated, and 6.2 % of its examples were truncated. The content signal is narrow: unrelated poems reach line-F1 0.553, and the teacher itself preserves little content. Author identity is measured automatically only.

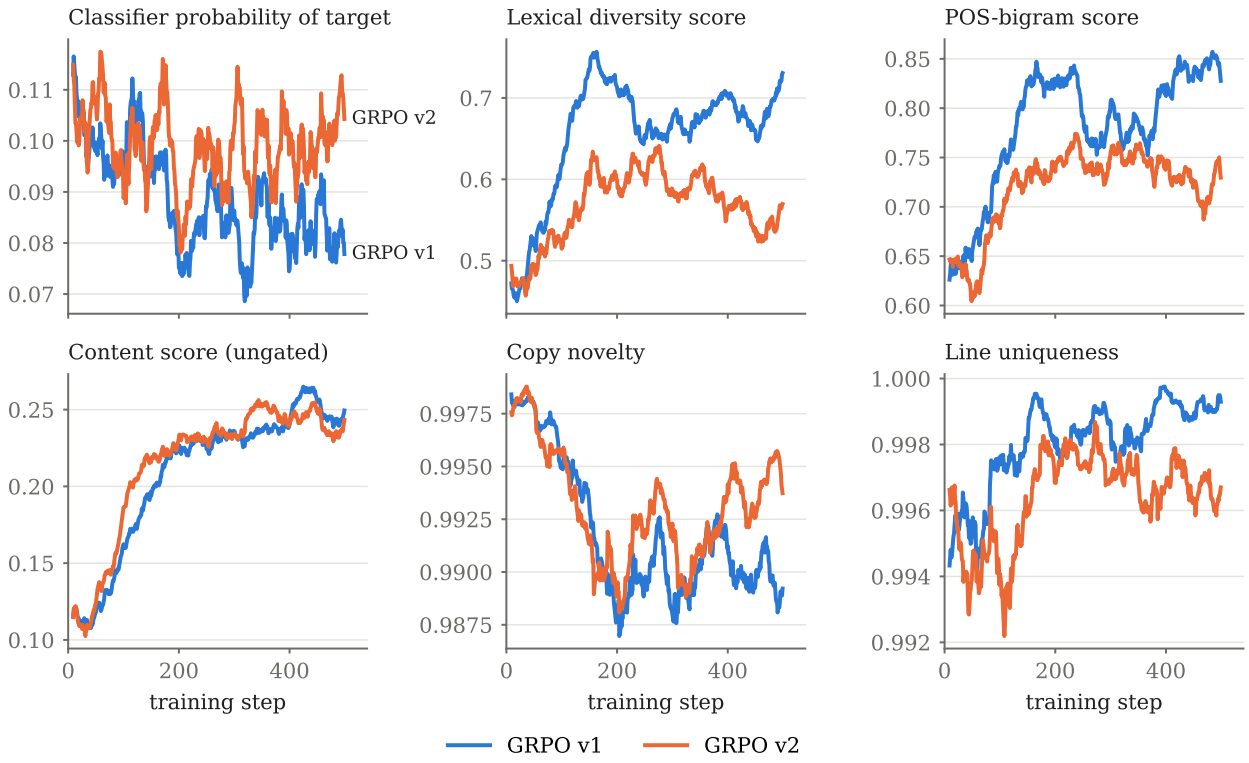


Figure 1: Reward components during training (mean over the 32 completions per step, 25-step moving average). Only components on a common scale are shown; R and the gated terms are not comparable between v1 and v2.

Reproducibility. Splits: `make_splits.py`. Teacher data: `generate_synthetic_parallel.sh`. SFT, GRPO and evaluation: `slurm/train_sft.slurm`, `train_grpo.slurm`, `train_grpo_v2.slurm`, `eval_grpo.slurm`. Tables: `compare_generations.py` with `--rewards` and `--corpus-reference` over the tracked generation CSVs in `data/validation_generations/`, which reproduces every number on CPU. Figure **??**: `scripts/plot_grpo_curves.py`. Every training run records its arguments, commit, merge provenance and library versions in `run_config.json`.